

Heat production problem solution

1. Estimate that rate of heat released by the people in this room in W.

Start with an energy balance for a single person

$$\begin{array}{ccccccc} \text{Energy accumulation} & = & \text{Energy} & - & \text{Heat energy} & - & \text{Work} & - & \text{Wasted energy} \\ \text{rate} & & \text{coming in} & & \text{loss} & & & & \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \text{Change in} & & \sim 14 \text{ MJ d}^{-1} & & \text{Most} & & \text{Not digested} \\ \text{body T} & & \text{in food.} & & \text{ends up} & & \\ \text{and fat etc.} & & \text{This is an} & & \text{as heat} & & \\ \text{storage. We} & & \text{average for} & & \text{anyway, so} & & \\ \text{will assume it is} & & \text{Europe.} & & \text{let's combine it} & & \\ \text{zero.} & & & & \text{with heat loss.} & & \end{array}$$

$$\begin{array}{ccc} \text{Heat energy} & = & \text{Food} & - & \text{Waste} \\ \text{loss} & & \text{energy in} & & \text{production} \\ & & & & \downarrow \text{guess 20\%?} \end{array}$$

$$\dot{Q}_{\text{out}} = \dot{Q}_{\text{food}} - 0.2 \dot{Q}_{\text{food}}$$

$$\dot{Q}_{\text{out}} = 0.8 \cdot 14 \text{ MJ/d} = 11.2 \text{ MJ/d}$$

$$= 11.2 \times 10^6 \text{ J/d} \cdot \frac{1 \text{ d}}{86400 \text{ s}} = 130 \text{ J/s} = 130 \text{ W}$$

There isn't much exertion in here.

Estimate $\sim 100 \text{ W}$ per person } interesting

~ 45 people in room so
 $\sim 4500 \text{ W}$ or $\sim 5 \text{ kW}$



Most of you were more familiar with energy in Calories. $2000 \text{ "Calories"/d} = 2 \cdot 10^6 \text{ calories/d} = 8.4 \text{ MJ/d}$.
2000 Calories is more like a minimum.

Conversion factor : 4.184 J/cal

Our estimate of 5 kW is only an estimate. Why might it be wrong?

My guess is it is accurate to $\pm 40\%$.